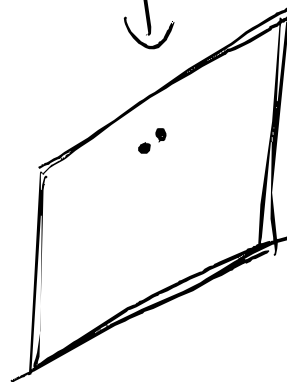
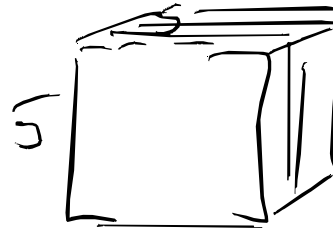
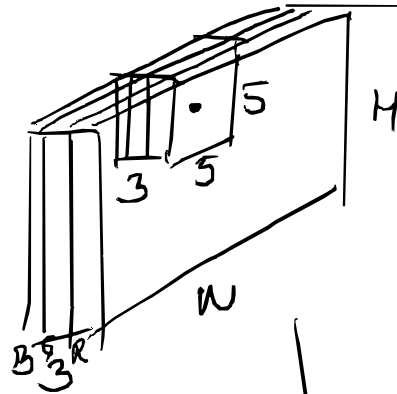


CNN

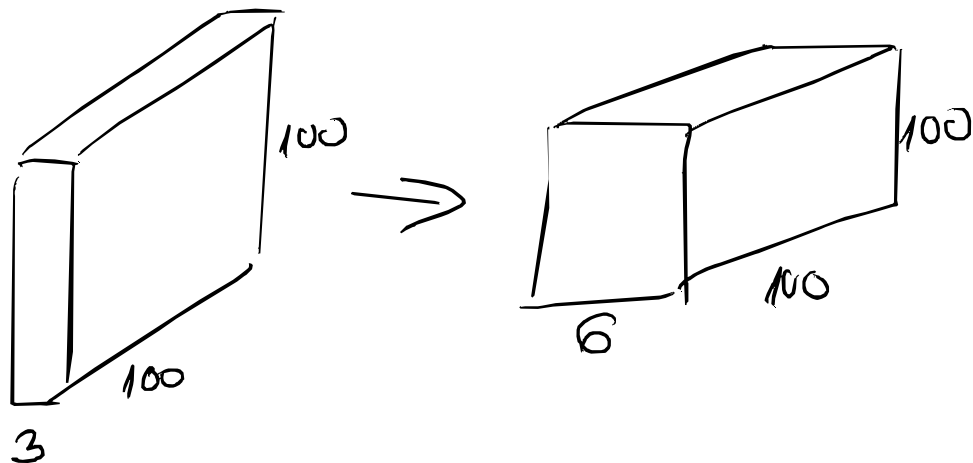
- KONVOLUCIJSKI SLOJ
- POOLING SLOJ (SLOJ ZDRAŽIVANJA)
- POTPUNO POVEZANI SLOJ

SKALARNI PRODUKT } OPERACIJE
MAX FUNKCIJA }

KONVOLUCIJA



AKTIVACIJSKA
MAPA $\Rightarrow$ REZULTAT
FILTRIRANJA 3D
FILTRUM



6 FILTERS
 $3 \times 3 \times \boxed{3}$
 PADDING 0A
 STRIDE (2,2)

$$\frac{N - F}{\text{stride}} + 1$$

$N = 100$
 $F = 3$
 stride 2
 zero padding $N = 102$

$$\frac{102 - 3}{2} + 1$$

~~99~~
~~2~~ we more

(2)

ZADATAK

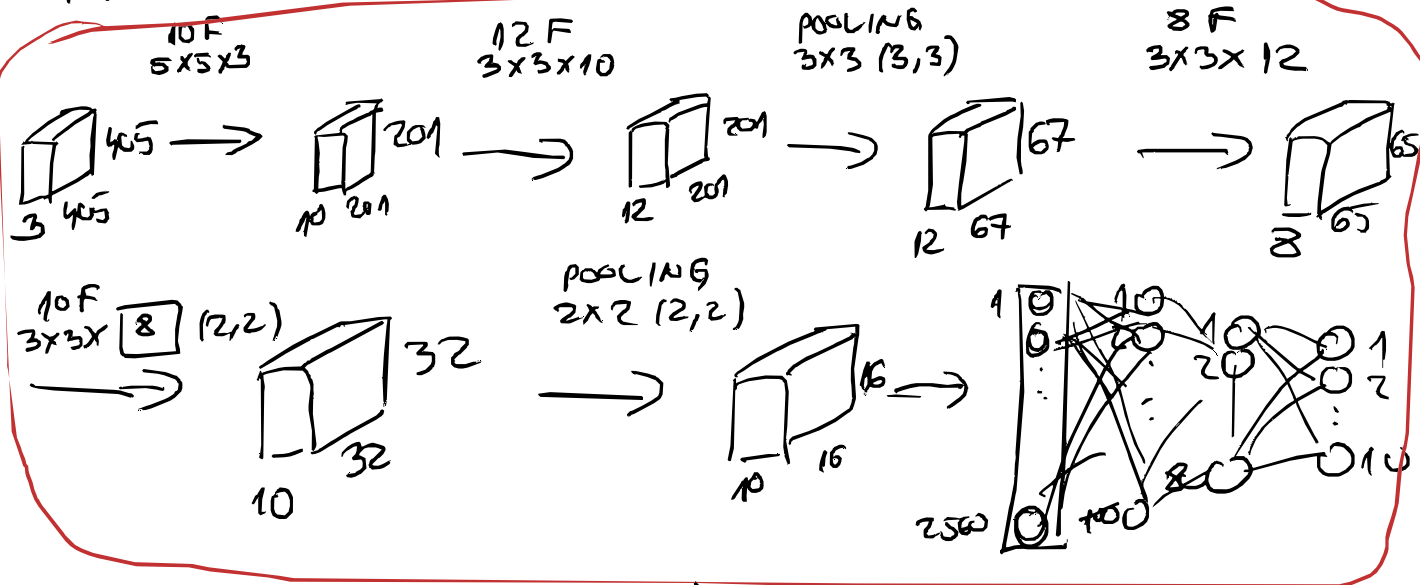
SKICIRAJ ARHITEKTURU SVJEDEČE MREŽE
I IZRAČUNAJ BROJ PARAMETARA MREŽE
AKO SU U KONVOLUCIJSKIM I FC
SLOJEVIMA UKLJUČENI BIASI.

- (1) KONV. SLOJ, 10 FILT. $5 \times 5 \times \square$, STRIDE (2,2)
- (2) KONV. SLOJ, 12 FILT $3 \times 3 \times \square$, PADDING
- (3) POOLING 3×3 , STRIDE (3,3)
- (4) KONV. SLOJ, 8 FILTERA $3 \times 3 \times \square$
- (5) KONV. SLOJ, 10 FILT. $3 \times 3 \times \square$, STRIDE (2,2)
- (6) POOLING 2×2 , STRIDE (2,2)

FLATTEN

- (7) FC SLOJ 100 NEURONA
- (8) FC SLOJ 80 NEURONA
- 10 KLASNA KLASIFIKACIJA $\Rightarrow$ 10 NEURONA

ULAZ JE SLIKA RGB REZOLUCIJE 405 X 405 PIKS



1. KONV. SLOJ $10 \cdot (5 \cdot 5 \cdot 3 + 1) = 760$ parametara
2. KONV. SLOJ $12 \cdot (3 \cdot 3 \cdot 10 + 1) = 1092$
3. KONV. SLOJ $8 \cdot (3 \cdot 3 \cdot 12 + 1) = 872$
4. KONV. SLOJ $10 \cdot (3 \cdot 3 \cdot 8 + 1) = 730$
- PRVI FC SLOJ $2560 \cdot 100 + 100 = 256100$
- DRUGI FC SLOJ $100 \cdot 80 + 80 = 8080$
- IZLAZNI $80 \cdot 10 = 800$

SUMA 268434 parametara

(DZ)

ULAZ $\Rightarrow$ RGB slika 200x200

- (1) KONV. SLOJ 10 FILT. 5x5x $\square$, stride (3,3)
- (2) POOLING 2x2, STRIDE (2,2)
- (3) KONV. SLOJ 6 FILT, 3x3x $\square$, PADDING
- FLATTEN
- (4) FC 100 NEURONA
- (5) FC 50 NEURONA
- (6) 10 KLASNA KLASIFIKACIJA

BIASI U SVIM SLOJEVIMA